

Beyond Error

Why Ensembles Work — Until They Don't

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Pre-Final Check-in



Recap: Motivation & Approach

Motivation

$$\text{MSE} = \text{Bias}^2 + \text{Variance} + \epsilon^2$$

Same drop in MSE can mean two very different things:

Bagging ↓ variance • **Boosting** ↓ bias

Approach

M=50 bootstrap samples → estimate Bias², Variance per config

Each config = one point in Bias²-Variance space

Vary 4 parameters → trace trajectories through that space

What we expected each parameter to do to Bias² and Variance

Parameter	Direction	Bagging Bias ²	Bagging Var	Boosting Bias ²	Boosting Var
Ensemble size	↑ more trees	→ flat	↓	↓	→ flat
Base depth	↑ deeper trees	↓	↑	↓	↑
Training size	↑ more data	↓	↓	↓	↓
Noise level	↑ more noise	→ stable	↑	↑	↑

Experimental Setup

Synthetic: 10,000 samples (8k train / 2k test) • 15 features

$$y = \sum w_i x_i + 3 \cdot \sin(2x_1) + 2 \cdot x_2^2 + 1.5 \cdot x_3 x_4 + \epsilon$$

Ensemble size $\in \{5, 10, 25, 50, 100, 200\}$ • Depth $\in \{2, 4, 6, 8, 10, 15, 20\}$ • Noise $\sigma \in \{5, 15, 30, 50, 75\}$ • Train size $\in \{500, 1k, 2k, 4k, 8k\}$

Bootstrap M=50 • Same bootstrap samples used for RF and XGB to enable covariance estimation

Hypotheses & Hybrid Formulations

Trajectory Hypotheses (H1–H4)

H1 More estimators

Adding more estimators moves RF mostly down (less variance) and XGB mostly left (less bias) — they don't follow the same path at all

H2 Boosting breaks with deep trees

Making trees deeper hurts XGB much more than RF — XGB's variance should explode while RF's stays controlled

H3 Boosting hungers for data

Adding more training data should shrink XGB's errors faster than RF's — boosting needs large samples to correct residuals reliably

H4 Noise breaks boosting, not bagging

Higher noise should hurt XGB in both bias and variance, while RF mainly picks up more variance — boosting tries to fit the noise signal

Hybrid Hypotheses (H5–H6)

H5 A mixture can beat both parents

There exists a weight λ where combining RF and XGB predictions achieves lower MSE than either model alone — possible when their errors are independent

H6 Deterministic blend beats random switch

Always mixing predictions (convex) should beat randomly picking one model per sample (probabilistic) — random switching adds extra variance from flipping between models

Hybrid Formulations

Convex Combination

$$f\lambda = \lambda \cdot f_{\text{XGB}} + (1-\lambda) \cdot f_{\text{RF}}$$

$$\text{Var} = \lambda^2 \cdot V_{\text{XGB}} + (1-\lambda)^2 \cdot V_{\text{RF}} + 2\lambda(1-\lambda) \cdot \text{Cov}(\text{RF}, \text{XGB})$$

→ always blends — no switching noise

Probabilistic Combination

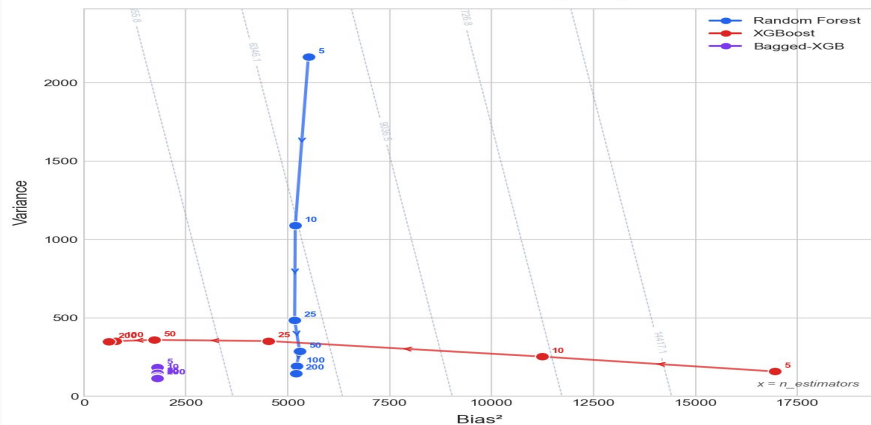
$$P(\text{pick XGB}) = \lambda, \quad P(\text{pick RF}) = 1-\lambda$$

$$\text{Var} = \lambda \cdot V_{\text{XGB}} + (1-\lambda) \cdot V_{\text{RF}} + \lambda(1-\lambda) \cdot (\mu_{\text{XGB}} - \mu_{\text{RF}})^2$$

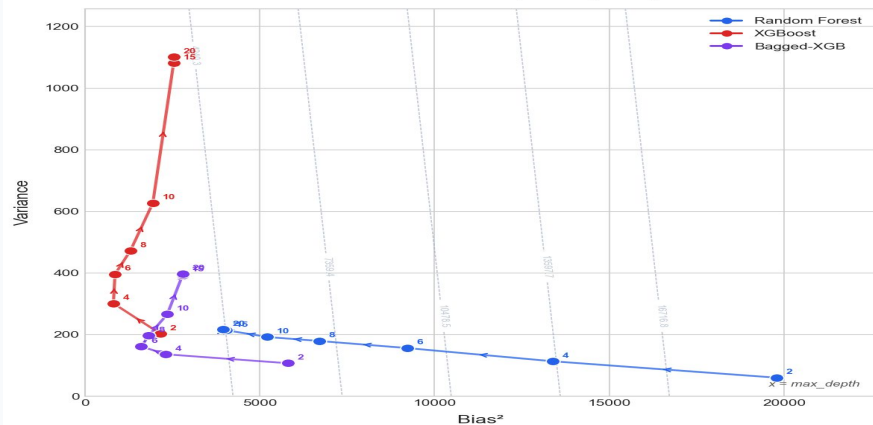
→ extra penalty for switching between models

Results: Trajectory Atlas

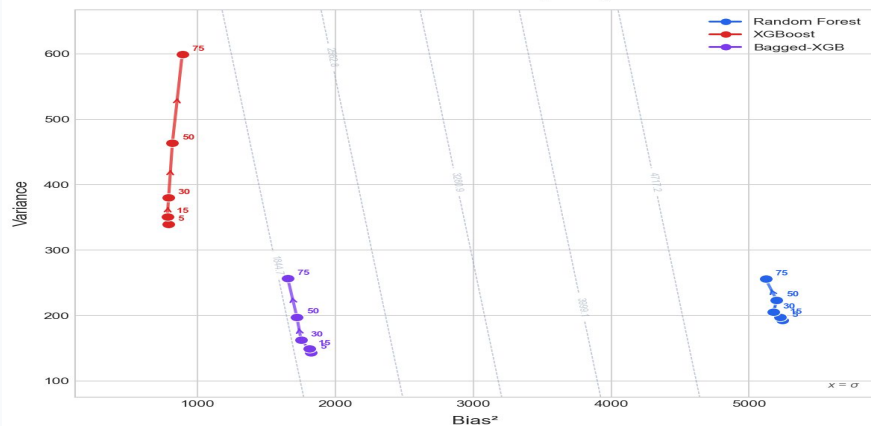
E1 · Ensemble Size Trajectory



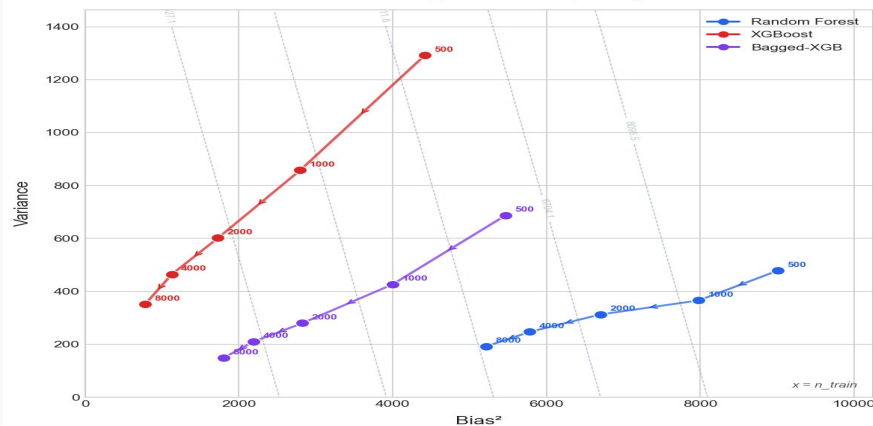
E2 · Base Learner Depth Trajectory



E3 · Noise Level Trajectory



E4 · Training Set Size Trajectory



Results: What the Trajectories Tell Us

The U-Turn: why XGBoost's bias increases at high depth

Boosting corrects errors by fitting residuals each round. Deep trees overfit those residuals — they learn noise rather than signal. Each subsequent round now amplifies noise, not corrects it. The model starts biasing toward noise patterns that don't generalise — so bias goes UP even as the model grows more complex. This is why XGBoost defaults to `max_depth = 6`.

Hypothesis Verdicts

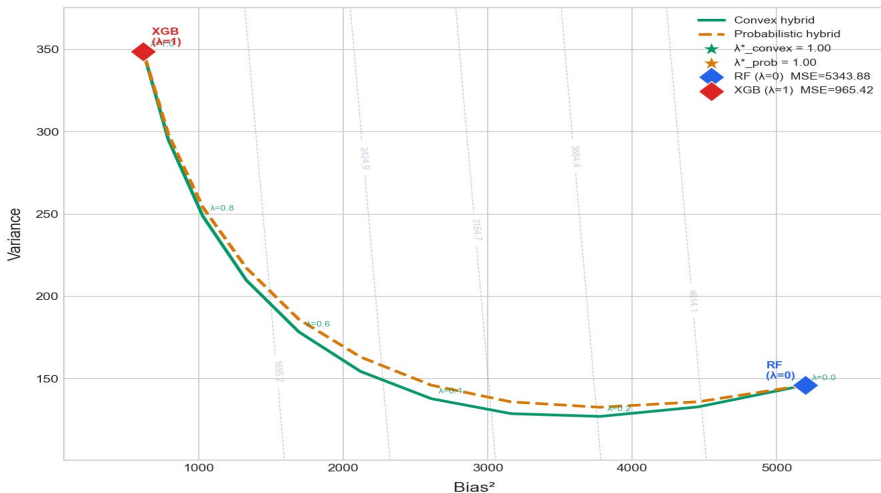
Parameter	Direction	Bagging Bias ²	Bagging Var	Boosting Bias ²	Boosting Var
Ensemble size	↑ more trees	→ flat	↓↓ sharply	↓ gradually	→ flat
Base depth	↑ deeper trees	↓ gradual	↑ moderate	↓ then ↑	↑↑ steep
Training size	↑ more data	↓	↓	↓↓ faster	↓↓ faster
Noise level	↑ more noise	→ stable	↑ moderate	↑ bias & var	↑↑ steep

Why Bagged-XGB sits in the sweet spot

Across all four experiments, Bagged-XGB (purple) clusters in the lower-left region, combining XGBoost's strength at reducing bias with bagging's ability to keep variance from exploding. Wrapping a boosting model in bagging acts as insurance: if individual XGBoost runs overfit differently, averaging them cancels the noise out.

Results: Hybrid Mixing & Why $\lambda^* = 1$

Plot 5 - Hybrid Curves in Bias²-Variance Space



Optimal λ depends on **bias gap + variance + covariance**.
In our case:

- RF bias $\gg$ XGB bias ($\sim 3x$)
- Even though RF var $<$ XGB Var - it is not enough to compensate
- This creates a **dominant positive term** in λ^*

Result:

“The optimal solution lies outside $[0,1]$, so it clips to $\lambda^* = 1$ ”

Convex Combination

$$f_{\lambda} = \lambda f_x + (1-\lambda) f_y$$

$$B_{\lambda} = \lambda B_x + (1-\lambda) B_y$$

$$V_{\lambda} = \lambda^2 V_x + (1-\lambda)^2 V_y + 2\lambda(1-\lambda) V_{xy}$$

$$\begin{aligned} \text{MSE}_{\lambda} &= B_{\lambda}^2 + V_{\lambda} \\ &= (\lambda B_x + (1-\lambda) B_y)^2 + \lambda^2 V_x + (1-\lambda)^2 V_y \\ &\quad + 2\lambda(1-\lambda) V_{xy} \end{aligned}$$

$$\frac{\partial \text{MSE}_{\lambda}}{\partial \lambda} = 0$$

$$\begin{aligned} \Rightarrow 2\lambda B_x^2 + 2(1-\lambda) B_y^2 + 2(1-2\lambda) B_x B_y \\ + 2\lambda V_x - 2(1-\lambda) V_y + 2(1-2\lambda) V_{xy} = 0 \end{aligned}$$

$$\begin{aligned} = \lambda (B_x^2 + B_y^2 - 2B_x B_y + V_x + V_y - 2V_{xy}) &= B_x^2 - B_x B_y \\ + V_x - V_{xy} \end{aligned}$$

$$\lambda_{\text{conv}}^* = \frac{(B_y - B_x) B_y + (V_x - V_{xy})}{(B_x - B_y)^2 + (V_x + V_y - 2V_{xy})}$$

Probabilistic Combination

$$\text{Bias}(f_{\lambda}) = \lambda B_x + (1-\lambda) B_y$$

$\Rightarrow$ Expectation is linear

$$\text{Var}(f_{\lambda}) = E[\text{Var}(f|\text{choice})] + \text{Var}[E(f|\text{choice})]$$

$\Rightarrow$ Law of total variance (Grev's law)

$$\lambda V_x + (1-\lambda) V_y$$

- linearity of exp

$$\lambda(1-\lambda) [E(f_x) - E(f_y)]^2$$

- var of Bernoulli style RV

$$p(1-p)(A-B)^2 = 0$$

Let $\Delta = E(f_x) - E(f_y)$ - gap b/w model predictions

$$\text{Var}(f_{\lambda}) = \lambda V_x + (1-\lambda) V_y + \lambda(1-\lambda) \Delta^2$$

$$\text{MSE}_{\lambda} = [\lambda B_x + (1-\lambda) B_y]^2 + \lambda V_x + (1-\lambda) V_y + \lambda(1-\lambda) \Delta^2$$

$$\begin{aligned} \frac{\partial \text{MSE}_{\lambda}}{\partial \lambda} &= 2[\lambda B_x + (1-\lambda) B_y](B_x - B_y) \\ &\quad + V_x - V_y + (1-2\lambda) \Delta^2 = 0 \end{aligned}$$

$$\lambda_{\text{prob}}^* = \frac{B_y(B_x - B_y) + V_x - V_y + \Delta^2/2}{(B_x - B_y)^2 + \Delta^2}$$

Conclusion & Future Work

What We Found

H1–H4 all confirmed: Bagging and boosting move in genuinely different directions

Boosting is fragile: Deep trees and high noise both cause bias AND variance to rise

Bagged-XGB works: Bagging around a booster keeps variance low without losing bias gains

Hybrids need comparable parents: With a 5.5× MSE gap, λ^* goes to 1 — just use the better model

Phase 2: What's Next

California Housing: Validate that trajectories hold on real-world noisy data

Heterogeneous ensembles: Mix structurally different models (SVM, Ridge, XGB)

Covariance analysis: Test if low $\text{Cov}(\text{RF}, \text{XGB})$ predicts when hybrids gain

Core Takeaway

MSE is a single number that hides the mechanism underneath. By tracking bias² and variance separately across parameter sweeps, we can see exactly how each ensemble method moves, and exactly where and why it breaks down. Boosting fails at high depth not because it gets 'too complex', but because it faithfully corrects the wrong signal: noise amplifies round by round.